

11.[Open Ended- SCAN Algorithm]

Consider a disk queue that contains requests for input/output operations at cylinder numbers **98, 183, 41, 122, 14, 124, 65, and 67**. The disk head is initially positioned at **cylinder 53**, and the cylinders are numbered from **0 to 199**. Use the **SCAN (Elevator) disk scheduling algorithm** to service these requests by moving the disk head in a particular direction and then reversing its direction after reaching the end. Write a program to determine the **total head movement**, measured in the number of cylinders traversed, while processing all the given requests

CODE :

```
GNU nano 8.7 scan.c Modified
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int total = 0;

    // Sorting
    for(int i=0;i<n-1;i++){
        for(int j=i+1;j<n;j++){
            if(req[i] > req[j]){
                int temp = req[i];
                req[i] = req[j];
                req[j] = temp;
            }
        }
    }

    int pos = 0;
    for(int i=0;i<n;i++){
        if(head < req[i]){
```

```
GNU nano 8.7 scan.c Modified
int pos = 0;
for(int i=0;i<n;i++){
    if(head < req[i]){
        pos = i;
        break;
    }
}

printf("Order of servicing:\n");

// Move RIGHT
for(int i=pos;i<n;i++){
    printf("%d -> ", req[i]);
    total += abs(head - req[i]);
    head = req[i];
}

// Go to end
total += abs(head - 199);
head = 199;

// Move LEFT
```

```
GNU nano 8.7 scan.c Modified
// Move RIGHT
for(int i=pos;i<n;i++){
    printf("%d -> ", req[i]);
    total += abs(head - req[i]);
    head = req[i];
}

// Go to end
total += abs(head - 199);
head = 199;

// Move LEFT
for(int i=pos-1;i>=0;i--){
    printf("%d -> ", req[i]);
    total += abs(head - req[i]);
    head = req[i];
}

printf("\nTotal Head Movement = %d\n", total);

return 0;
}
```

OUTPUT :

```
ASUS@Kalash-Laptop MINGW64 ~
$ nano scan.c

ASUS@Kalash-Laptop MINGW64 ~
$ gcc scan.c -o scan

ASUS@Kalash-Laptop MINGW64 ~
$ ./scan
Order of servicing:
65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 41 -> 14 ->
Total Head Movement = 331

ASUS@Kalash-Laptop MINGW64 ~
$
```